



UNIVERSITY OF PISA · SUMMER SCHOOL 2026

Text Analysis and Large Language Models for Innovation Management

1 – 4 September 2026 · Welcome and Introduction



UNIVERSITÀ DI PISA



BUSINESS ENGINEERING FOR DATA SCIENCE

Welcome to the Summer School

WHAT

An intensive learning experience at the intersection of text analysis, NLP, large language models and innovation management.

WHO

PhD students, researchers and scholars working with unstructured text: patents, scientific publications, reviews, business and organisational documents.

HOW

Methodological lectures, hands-on sessions, expert keynotes and collaborative project work — with your own research at the centre.

What you will be able to do

1

Use the core concepts and terminology of text analysis, NLP and LLMs

2

Identify textual sources suited to innovation and management research

3

Select methods that are consistent with your research objectives

4

Implement foundational and advanced NLP techniques

5

Use large language models critically and responsibly

6

Assess validity, reliability, reproducibility and interpretability

7

Recognise the limitations, risks and biases of computational methods

8

Apply all of it to your own individual or collaborative project

Agenda

	DAY 1 1 September	DAY 2 2 September	DAY 3 3 September	DAY 4 4 September
9.00 – 11.00	Welcome and Introduction	Text Similarity	Text Classification	Vito Giordano Rigorous Text Analysis in Innovation Research
11.00 – 13.00	Text Pre-processing	Meet the Editors R&D Management · Technovation	Large Language Models	Jane Lê AI in Research Methods: An Innovation Too Far? Stefan Feuerriegel A Consensus-Based Reporting Checklist for LLMs in Behavioral and Social Science
13.00 – 14.00	Lunch			
14.00 – 15.30	Topic Modelling	Project Work	Project Work	Project Work
15.30 – 17.00	Project Work	Ivan Zupic A Human-Centered Workflow for Using Large Language Models in Content Analysis	Andrea Bonaccorsi From Text to Action: Challenges for Innovation, Management, and Society	Project Work Presentation
20.00	Social Aperitif		Social Dinner	

Full programme, slides, notebooks and datasets: github.com/VitoGiordano94/Summer-School---Text-Analysis-and-Large-Language-Models-for-Innovation-Management

Everything is on GitHub

github.com/VitoGiordano94/Summer-School---Text-Analysis-and-Large-Language-Models-for-Innovation-Management

Lessons

Slides and methodological materials for every session.

Code

Notebooks and scripts, ready to run and to adapt.

Exercises

Guided practice on real innovation datasets.

Keynotes

Materials from the invited expert talks.

The repository stays useful after the school. Adapt the notebooks to your own research questions — and keep asking what each technique assumes, and what it cannot see.

Telegram Channel



Icebreaker: Human Topic Modelling

You are documents. Discover the main research topics represented in this room , and meet the people behind them.

Build your document

You need

3 MINUTES

- 1 Your name badge
- 2 One A4 sheet
- 3 One pen

Hold the sheet vertically and split it into two sections: what you study, and the kind of text you work with.

Maximum 5–8 words per section. Write in LARGE CAPITAL LETTERS, so that others can read your sheet from across the room.

I STUDY...

**GREEN
TECHNOLOGY
ADOPTION**

MY TEXTS ARE...

PATENT ABSTRACTS

Step 1 – Silent exploration

Stand up, move out, hold your sheet in front of you, and read the other documents.

5 MINUTES

1

Do not speak

Reading only — this is the feature-extraction phase.

2

Look for similarities

Both in research interests and in types of text.

3

Use gestures

If you spot a possible connection, signal it.

4

Keep moving

You do not need to decide anything yet.

Step 2 – Check the similarity

Now you can speak. Talk briefly with at least three different people.

5 MINUTES

What is the connection
between our research?

Are our texts similar or
different?

Could we belong to the same
topic?

Then form a cluster of 3 to 6 people.

There is no single correct classification. Your task is to find a connection that is meaningful and interesting.

Step 3 – Meet your cluster

In your cluster, each participant has about 45 seconds.

5 MINUTES

- 1 Your name and institution
- 2 What you study
- 3 How you currently use NLP, LLMs or other language technologies
- 4 One thing you hope to learn during the Summer School

Listen for shared interests, different experiences and possible collaborations.

Step 4 – Interpret your topic

As a group, agree on four things.

4 MINUTES

1

Your topic label

A short, creative name.

Emerging Technology Detectives · Patent and
Science Miners · Markets Through Text

2

Your shared connection

What brings your research projects together?

3

Your shared expectation

What would several members of the group like to
learn here?

4

Your bridge document

Who brings the least obvious — but most interesting
— connection to the group?

Step 5 – Present your topic

One spokesperson per cluster. Maximum 30 seconds.

4–5 MINUTES

“We are the topic called _____.
We are connected by _____.
We hope to learn more about _____.
Our bridge document is _____, because _____.”

Keep it to 30 seconds — we want every topic in the room to be heard.

What just happened?

We just built a human topic model — and then broke it open.

TWO VISIBLE FEATURES

- 1 What we study
- 2 The texts we analyse

WHAT CONVERSATION REVEALED

- 1 Experience with language technologies
- 2 Learning expectations
- 3 Unexpected research connections
- 4 Opportunities for collaboration

The same documents can produce different topics.

It depends on the question we ask — which is exactly the problem we spend the next four days on.

- **Definitions (in brief)**
- Why today?
- A Framework

Definitions

Natural Language Processing (NLP): extract a fuller meaning mathematical representation of unstructured text

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Text Mining (TM): extraction of non-trivial knowledge from unstructured text

Generative Large Language Models: Deep Learning models created with NLP and TM techniques, that are able to perform multiple text-related tasks

Definitions

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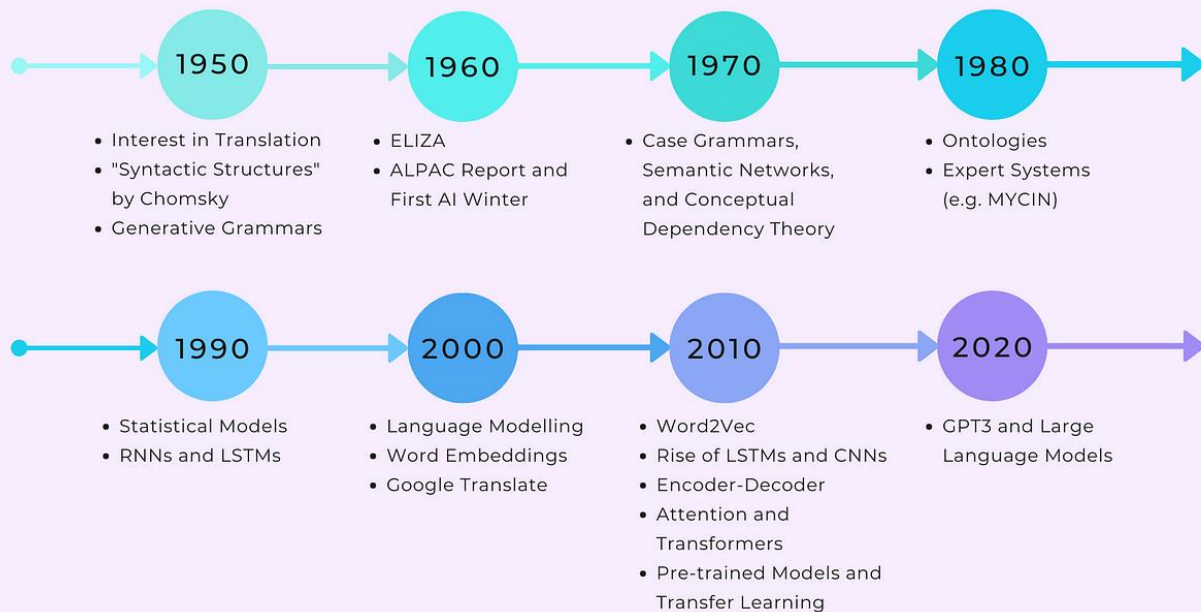
Text Mining (TM): extraction of non-trivial knowledge from unstructured text

Generative Large Language Models: Deep Learning models created with NLP and TM techniques, that are able to perform multiple text-related tasks

Other words used as synonyms: Computational Text Analysis, Linguistic Data Processing, Automated Text Analysis, Textual Data Analysis, Computational Linguistics, Text Data Mining, Computational Language Analysis...

History of NLP

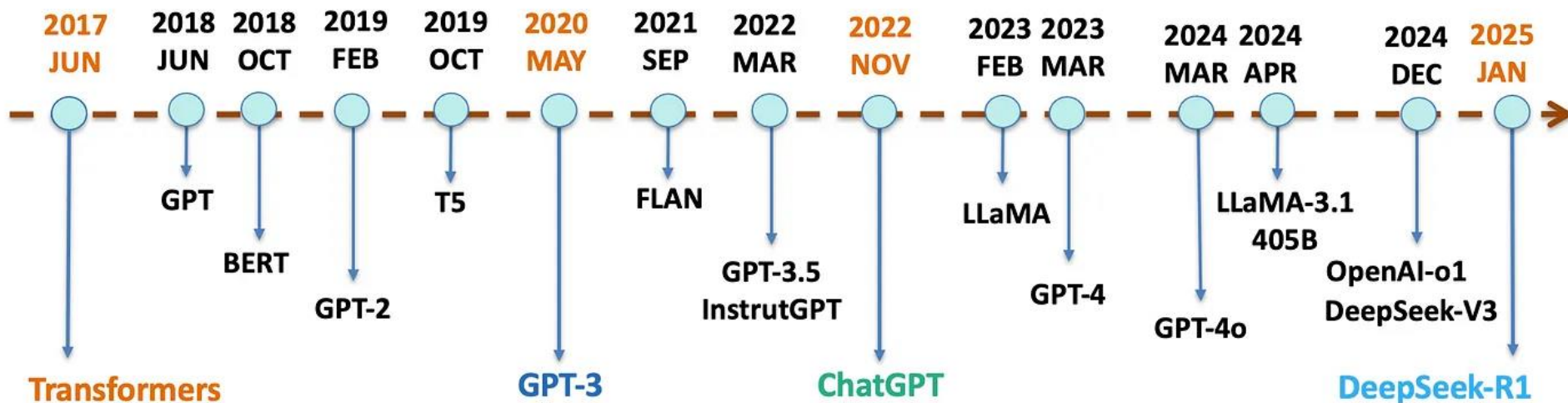
A Brief Timeline of NLP



Fabio Cusano (2023), A Brief Timeline of NLP, Medium

History of Generative LLMs

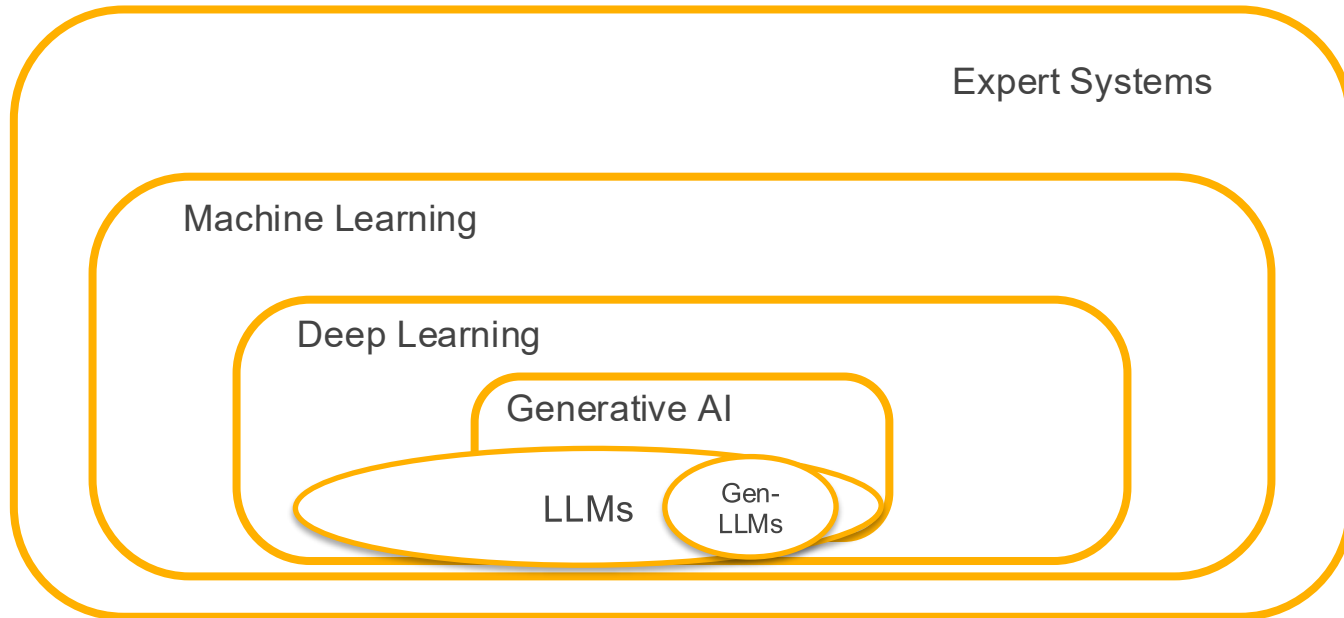
A Brief History of LLMs



Gilles Legoux (2023), History of the Generative AI, , Medium

LLMS

Artificial Intelligence

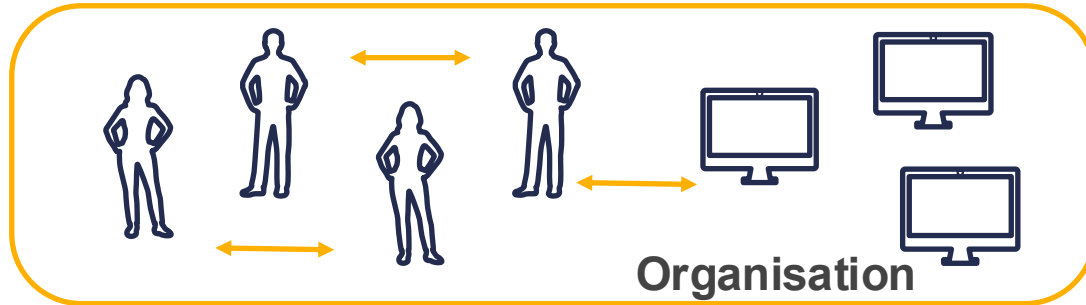


- Definitions (in brief)
- **Why today?**
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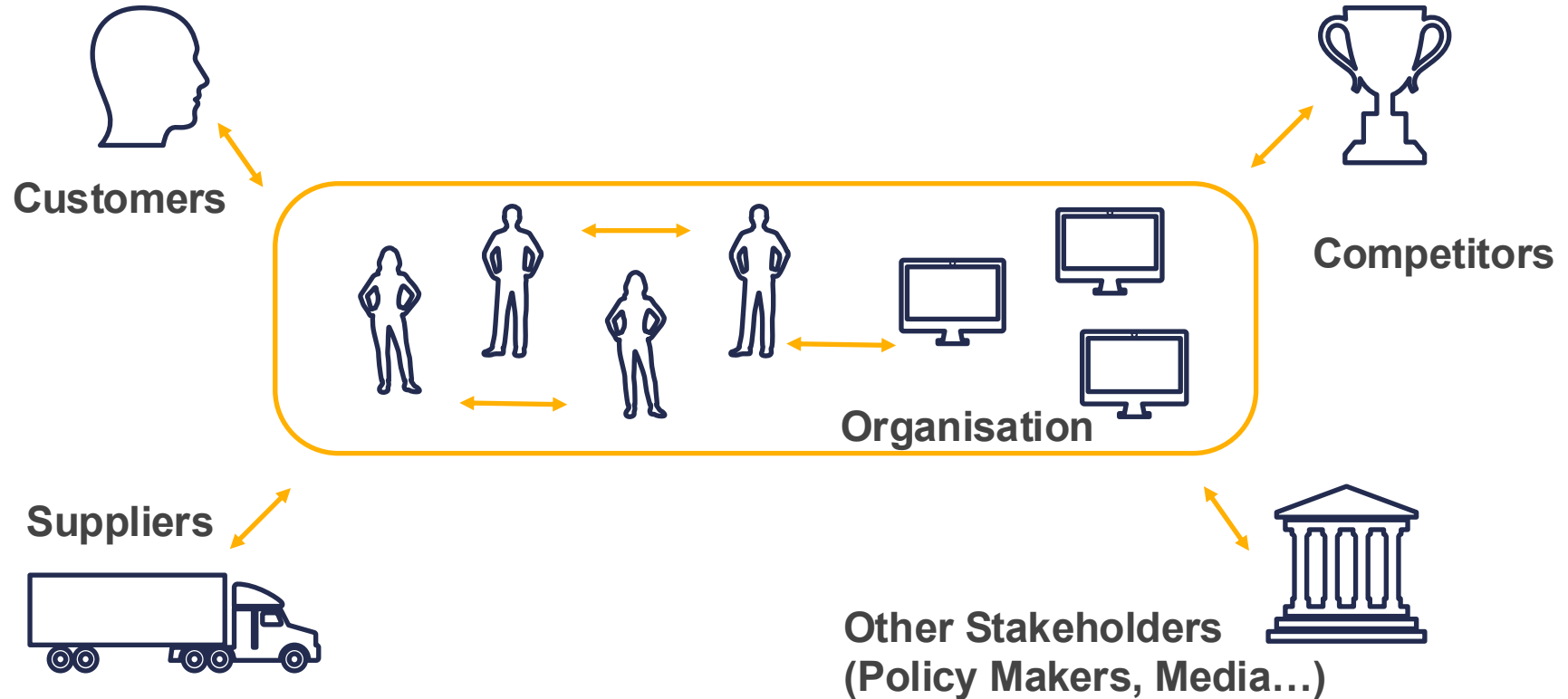
Text is everywhere

Around 80% of data in organizations are commonly estimated to consist of unstructured text. (Grimes, 2008).

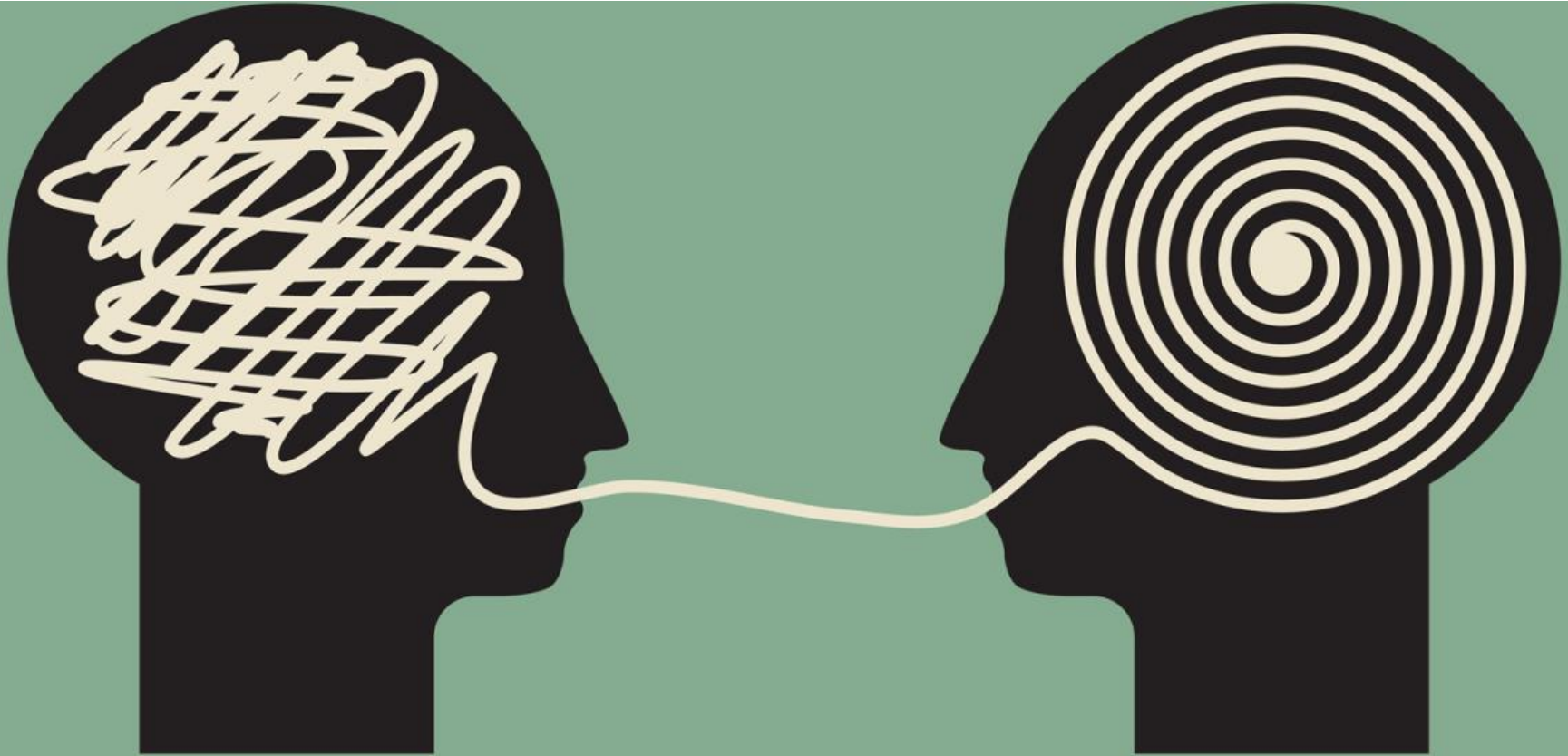
Text is everywhere



Text is everywhere



Text is us



Text is them



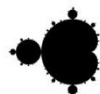
Text is increasingly accessible



spaCy



AllenNLP



TextBlob



- Definitions (in brief)
- Why today?
- **A Framework**

Innovate in Methods to Study Innovation*

We see a trend towards applying text mining techniques to:

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- explore large quantity of documents

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We see a trend towards applying text mining techniques to:

- explore large quantity of documents
 - bridge qualitative and quantitative research traditions by structuring, analyzing, and understanding text
 - operationalize new, previously unexplored variables
 - improve the measurement of existing variables
-
- Wetzels, M., Wetzels, R., Schweiger, E., & Grewal, D. (2025, June 7). How topic modeling can spur innovation management. *Journal of Product Innovation Management*.
 - Antons, D., Grünwald, E., Cichy, P., & Salge, T. O. (2020). The application of text mining methods in innovation research: current state, evolution patterns, and development priorities. *R&D Management*, 50(3), 329-351. -
 - Nambisan, S., Lyytinen, K., Majchrzak, A., & Song, M. (2017). Digital innovation management. *MIS quarterly*, 41(1), 223-238.

Innovate in Methods to Study Innovation*

Text mining allows management scholar's not only to harness the value of larger bodies of textual data...

Innovate in Methods to Study Innovation*

Text mining allows management scholar's not only to harness the value of larger bodies of textual data...

but also to increase the **objectivity and reproducibility** of empirical findings.

How?

R&D
MANAGEMENT



The application of text mining methods in innovation research: current state, evolution patterns, and development priorities

David Antons^a, Eduard Grünwald, Patrick Cichy and Torsten Oliver Salge

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CATALYST

Augmenting human innovation teams with artificial intelligence: Exploring transformer-based language models

Sebastian G. Bouschery¹ | Vera Blazevic^{1,2} | Frank T. Piller¹

Technovation 129 (2024) 102883



Contents lists available at ScienceDirect

Technovation

journal homepage: www.elsevier.com/locate/technovation



Natural language processing for innovation search – Reviewing an emerging non-human innovation intermediary

Julian Just

University of Innsbruck, Team Innovation & Entrepreneurship, Universitätsstraße 15, 6020, Innsbruck, Austria

JOURNAL OF PRODUCT
INNOVATION MANAGEMENT



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Future applications of generative large language models: A data-driven case study on ChatGPT

Filippo Chiarello^{a,d,*}, Vito Giordano^{a,d}, Irene Spada^{b,d}, Simone Barandoni^{c,d}, Gualtiero Fantoni^{b,d}

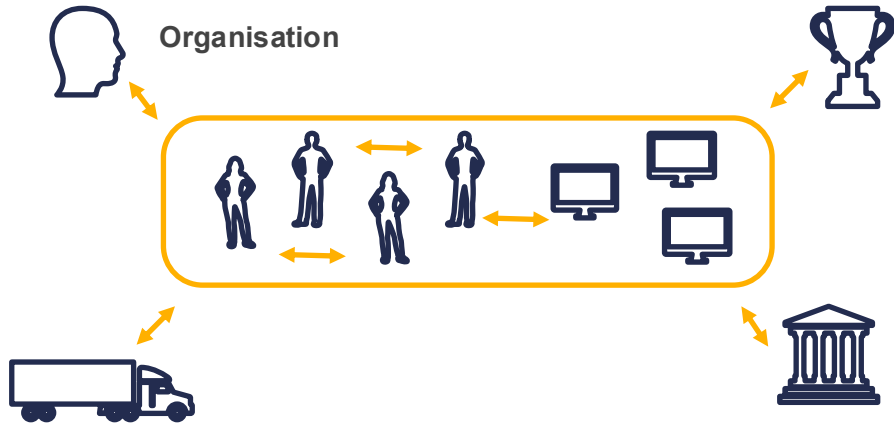
^a Department of Energy, Systems, Territory and Construction Engineering, University of Pisa, Italy

^b Department of Civil and Industrial Engineering, University of Pisa, Italy

^c Department of Computer Science, University of Pisa, Italy

^d Business Engineering for Data Science (B4DS) Research Lab, University of Pisa, Italy

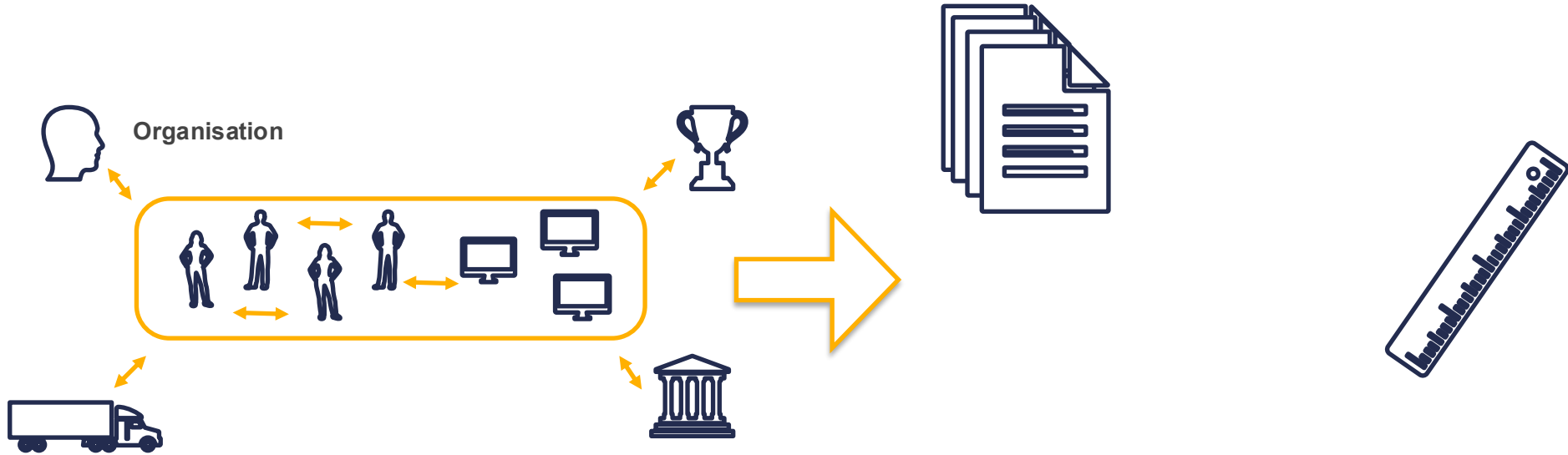
Measuring Texts in Management



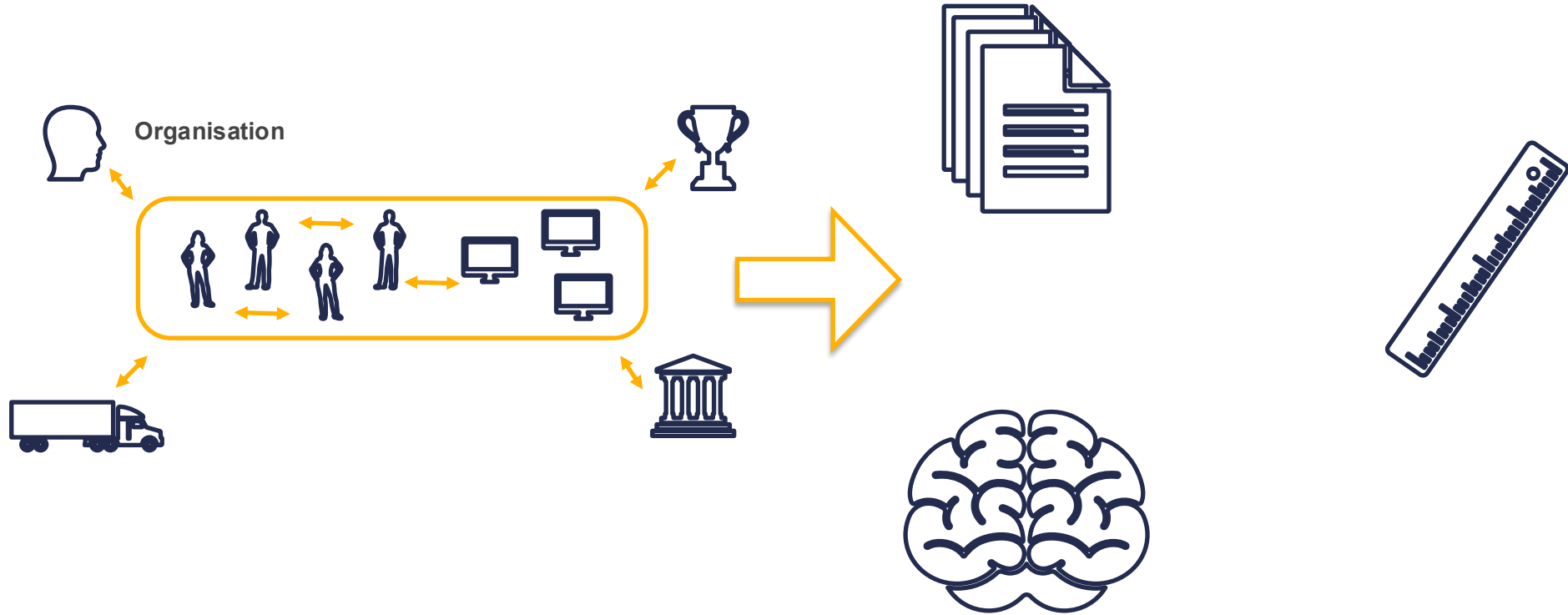
Measuring Texts in Management



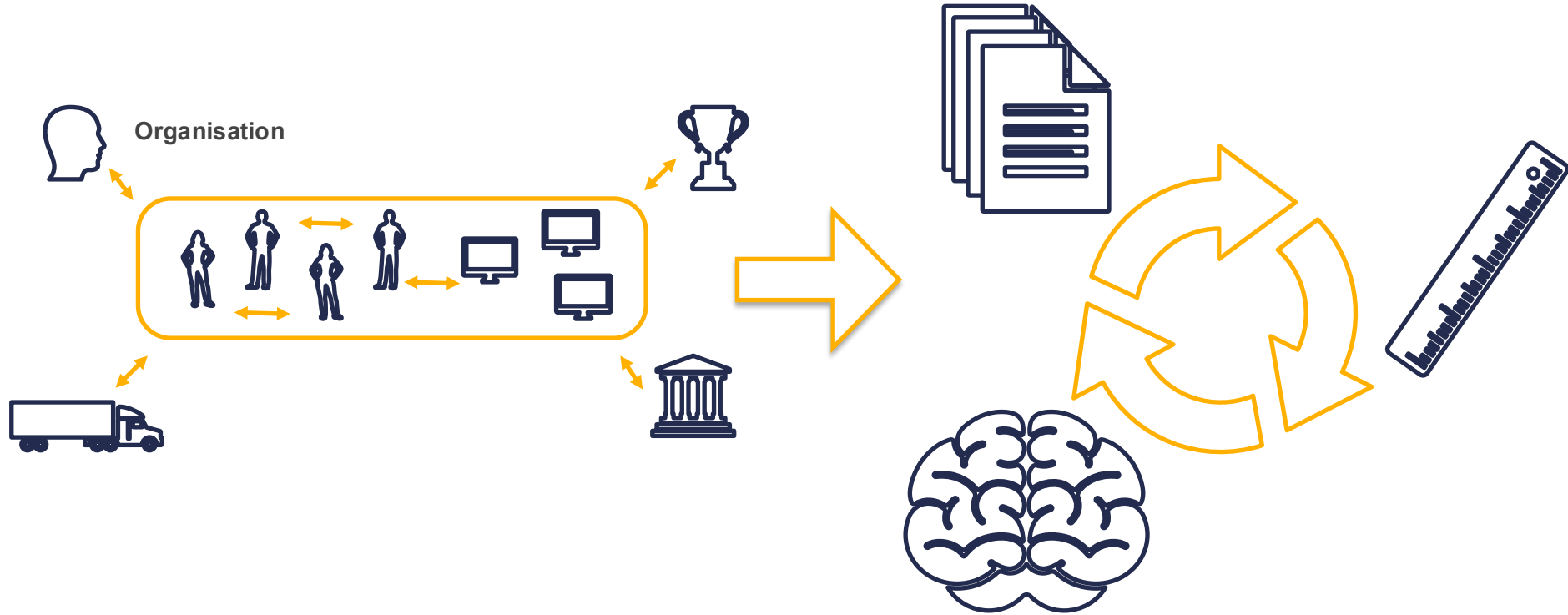
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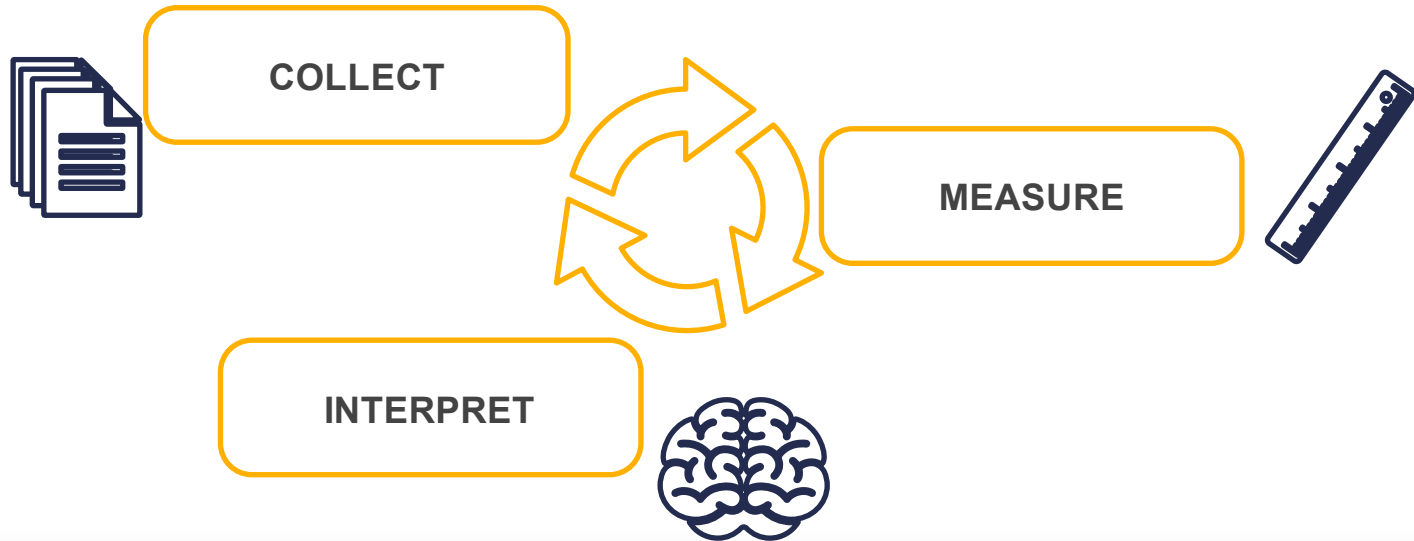


Measuring Texts in Management

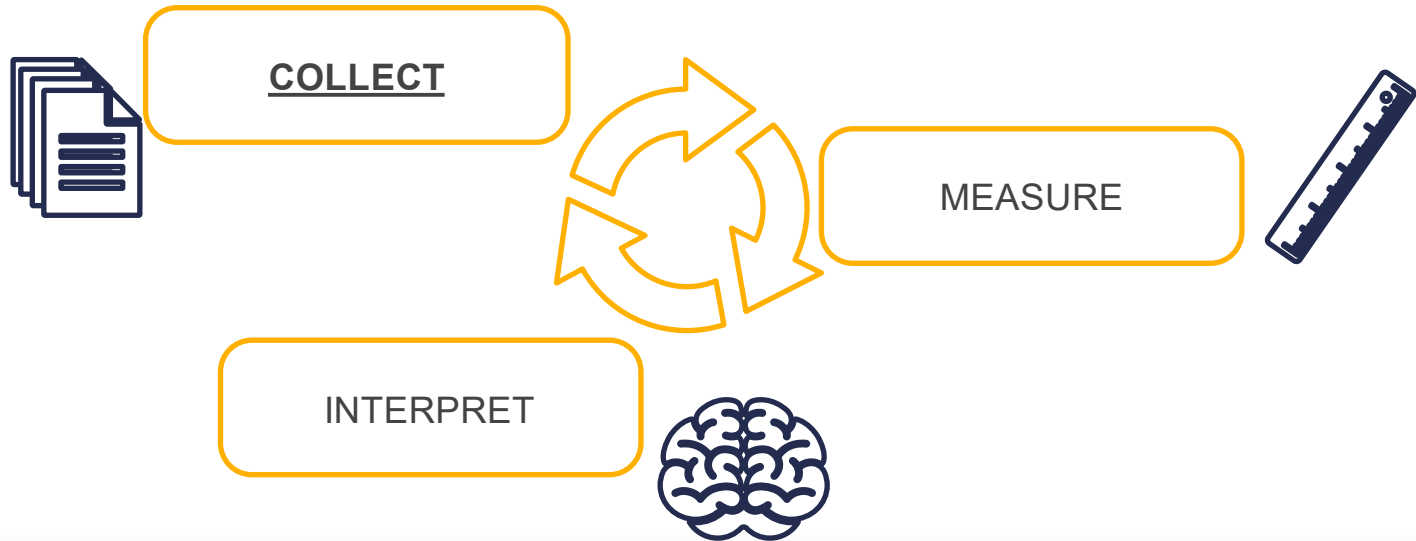


Measuring Texts in Management

To do so, researchers has to create their tool-box of instruments, that can enable to understand what is measurable when considering texts in (and outside) organizations. The toolbox should enable to **(1) collect** these texts, **(2) measure** them and **(3) interpret** the results.



Measuring Texts in Management



Data Collection Techniques

1. Search Strategy Design

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2. Web Scraping

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3. Collaboration with Companies

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1. Search Strategy Design

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3. Collaboration with Companies

4. Questionnaires & Interviews

Collect: some dimensions for reflecting

Level of Analysis

- Macro Level: Industry reports, National or global economic policies, Corporate annual report
- Meso Level: Project reports, CVs, Internal newsletters, Inter-departmental emails
- Micro Level: Individual emails, memos, Meeting transcripts, Employee feedback forms

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- Internal Sources: Emails, meeting minutes, reports, internal chat logs, policy documents.
- External Sources: Social media posts, news articles, customer reviews, academic publications, websites.

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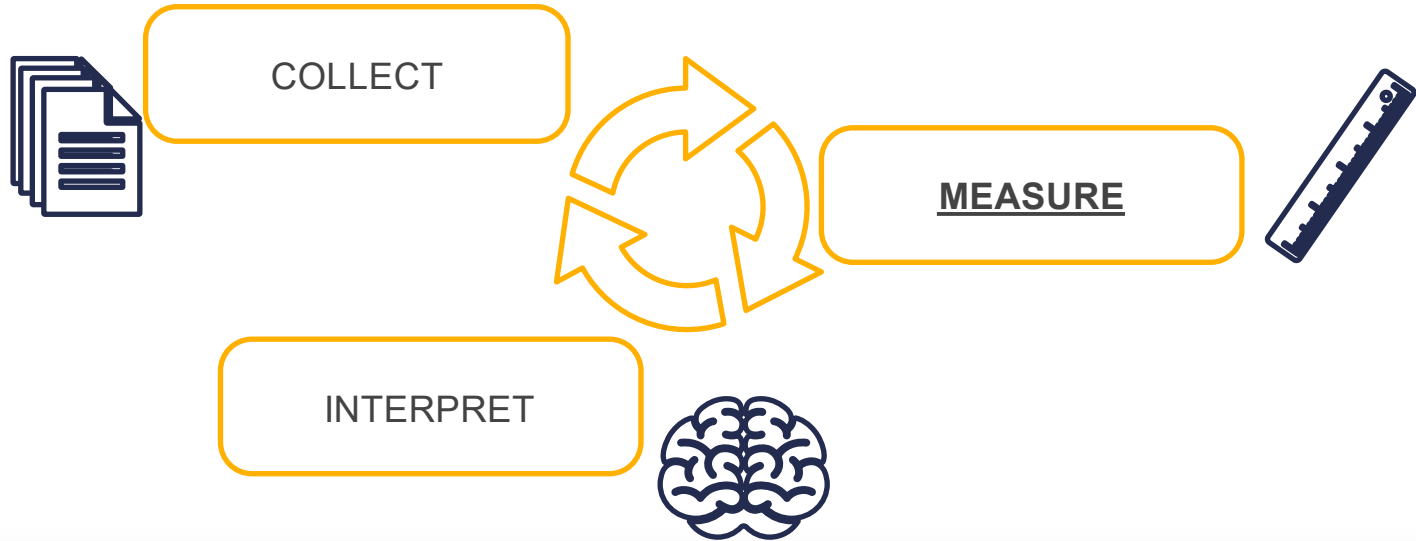
Data Sources:

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Data Collection Methods:

- Web Scraping Tools: BeautifulSoup, Scrapy, Selenium.
- APIs: Twitter API, Facebook Graph API, Google News API.
- Databases: Accessing organizational databases and document management systems.
- Data Gathering: Questionnaires, focus groups

Measuring Texts in Management



Measure

Measure: to discover the exact size or amount of something

How to make unstructured text measurable?

Natural Language Processing (NLP): extract a fuller meaning mathematical representation of unstructured text

What is measurable in texts?

Text Mining (TM): extraction of non-trivial knowledge from unstructured text

+ Generative LLMs

NLP Techniques

- **Tokenization**: Splitting text into individual words, phrases, or sentences.
- **Part-of-Speech (POS) Tagging**: Assigning parts of speech (e.g., nouns, verbs) to each word in a sentence.
- **Named Entity Recognition (NER)**: Identifying and classifying entities (e.g., names, dates, locations) in text.
- **Lemmatization**: Reducing words to their base or root form.
- **Stemming**: Reducing words to their root forms by removing suffixes.
- **Dependency Parsing**: Analyzing the grammatical structure of a sentence and establishing relationships between words.
- **Word Embeddings**: Representing words in a continuous vector space (e.g. BoW, Word2Vec, GloVe, BERT, GPT).

TM Techniques

- **Text Classification:** Assigning predefined categories to text (e.g., spam detection, sentiment analysis, topic labeling). Techniques: Naive Bayes, Support Vector Machines (SVM), Random Forests.
- **Sentiment Analysis:** Determining the emotional tone (positive, negative, neutral) of text.
- **Text Clustering:** Grouping similar documents together based on their content. K-Means, Hierarchical Clustering.
- **Topic Modeling:** Identifying hidden themes or topics within a set of texts. Techniques: Latent Dirichlet Allocation (LDA), Non-Negative Matrix Factorization (NMF).
- **Association Rule Mining:** Discovering interesting relations between variables in large datasets (e.g., market basket analysis).
- **Keyword Extraction:** Identifying the most important words or phrases in a text.
- **Co-occurrence Network Analysis:** Analyzing the co-occurrence of words or entities within a text to understand relationships.
- **Document Similarity:** Measuring the similarity between texts. Techniques: Cosine Similarity,

GenLLMs Techniques

Almost all the previous and...

- **Language Modeling:** Predicting the next word in a sequence of words. Generating text that continues a given prompt, producing contextually relevant and coherent sentences.
- **Text Summarization:** Creating a condensed version of a text while retaining key information:
- **Machine Translation:** Translating text from one language to another

Measure: the output (Document Level)

	Metadata			Document Features		
Document ID	<i>year</i>	<i>author</i>	<i>...</i>	<i>Topic</i>	<i>Sentiment</i>	<i>....</i>
1						
2						
3						
...						

Measure: the output (Document Level)

What else?

Numerical:

- *Sentiment Score*: Quantifies the overall sentiment of the text, typically on a scale from negative to positive.
- *Classification Probability*: The likelihood that a document belongs to a particular category or class.
- *Topic Probability*: The likelihood that a document pertains to a specific topic.
- *Number of Words Belonging to a Lexicon*: Counts the occurrence of words from a predefined lexicon within the document.

Categorical:

- *Classification Output*: The assigned category or class of the document, such as spam/not spam, topic labels, etc.
- *Topic*: The dominant topic or themes identified in the document.
- *Yes/No Answer Given by Generative LLMs*: Binary outputs generated by language models in response to queries.
- *Named Entities*: Entities such as persons, organizations, locations, identified in the text.

Measure: the output (Word Level)

		Metadata			Word Feature		
Document ID	Word ID	<i>e.g. year</i>	<i>e.g. author</i>	...	<i>Entity Class</i>	<i>Part-of-speech</i>	...
1	1						
1	2						
1	3						
...	..						

Measure: the output (Word Level)

Numerical:

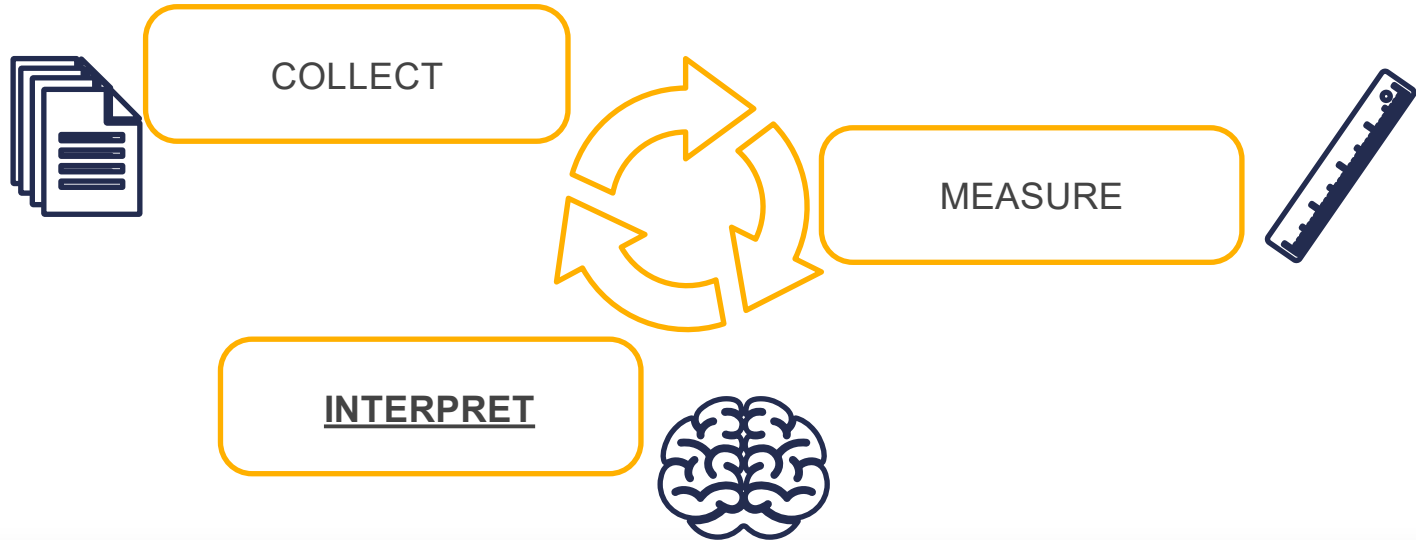
- *Importance of a Word in a Document:* Metrics like TF-IDF score indicating the significance of a word within a document relative to a corpus.
- *Level of Abstraction of a Word:* A score indicating how abstract or concrete a word is, often derived from lexical databases.
- *Complexity of a Word:* Measured by factors such as syllable count, word length, or frequency of use in the language.
- *Word Embedding Vector:* A high-dimensional vector representation of a word used in NLP models, capturing semantic relationships.

Measure: the output (Word Level)

Categorical:

- Word Belongs to a Lexicon (Y/N): Indicates whether the word is part of a predefined lexicon or dictionary.
- Part of Speech: Classification of the word as noun, verb, adjective, etc.
- Named Entity Recognition: Identification of the word as a named entity like person, organization, location, etc.
- Domain-Specific Classification: Identification of the word as belonging to a specific domain, such as medical, legal, technical.

Measuring Texts in Management



Interpret

Prior to knowledge generation, the validity of TM- based findings will need to be established:

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- When TM is used to identify and operationalize constructs, using different forms of data triangulation will help generate construct validity evidence.

Interpret

Prior to knowledge generation, the validity of TM- based findings will need to be established:

- Learn to use standard measures: precision, recall, F1-score
- When TM is used to identify and operationalize constructs, using different forms of data triangulation will help generate construct validity evidence.
- TM need comparative evaluation in which (part of) the TM output is correlated with independent data sources or other “standards” (such as the aforementioned survey or expert data).

Descriptive Research

To describe characteristics of a population or phenomenon.

Content Analysis: Extracting and quantifying themes, keywords, and phrases from large text datasets to describe trends.

Frequency Analysis: Counting word occurrences to identify the most common terms or concepts in a text corpus.

Graph analysis: measuring relations between words and plot them as a graph



Exploratory Research

To explore a phenomenon without predefined hypotheses.

Pattern Discovery: Using unsupervised learning techniques (e.g., clustering, topic modeling) to uncover unknown patterns and themes.

Trend Analysis: Identifying emerging trends and shifts in text data over time.



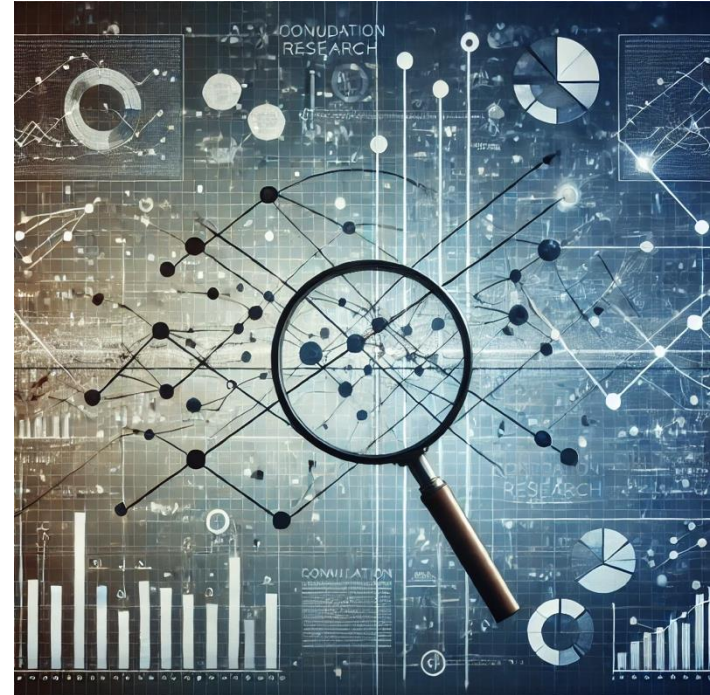
Correlation Research

To identify relationships between variables without inferring causality.

Co-occurrence Analysis:

Investigating how often terms or concepts appear together in texts, indicating potential relationships.

Topic Modeling: Exploring the correlation between different topics discussed in documents.

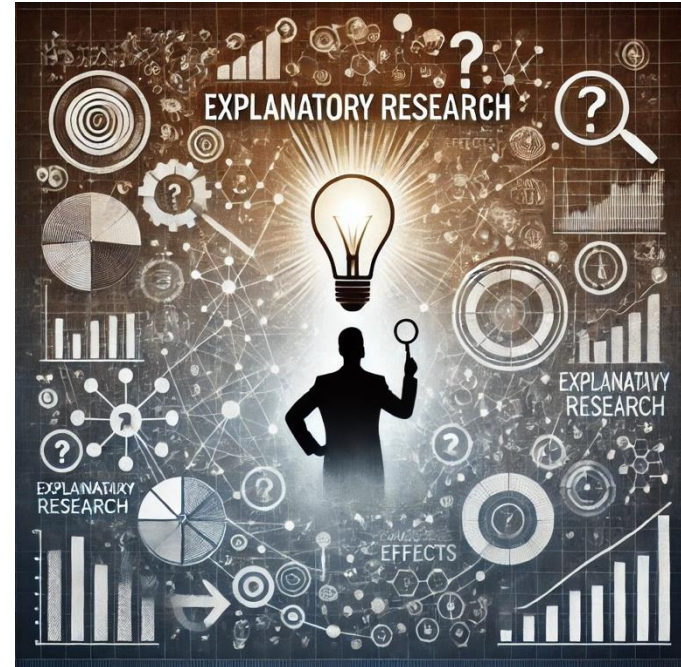


Explanatory Research

To explain the reasons behind a phenomenon.

Causal Inference: Using text data to support hypotheses about causal relationships (e.g., using sentiment analysis to explain consumer behavior).

Regression Analysis: Employing text-derived variables in regression models to explain dependent variables.



A Call to Action

Text Mining can become the telescope of management studies



Thank you for your attention.

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